Name
Date
Course

Beginning Algebra Practice Final Test (Part II) Version 2

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show work, even when you can do the problems in your head.
iii) Draw a rectangle around your answers.
iv) Staple a copy of these test questions with your work and your final answers.
v) You may use a calculator (but not a phone or a graphing calculator).

1) Find an equation of the line through the points $(3,7)$ and $(-2,2)$ in

a) Point-Slope form.

b) Slope-intercept form.

c) What is the slope of the line?

d) What is the y-intercept?

2) Use the slope-intercept form to find the slope-intercept form of the equation with slope $\frac{2}{3}$ and point $(-1,6)$ on the line.

3) Find an equation of the line with the following characteristics.

x-intercept: 2
y-intercept: -4

④ Find an equation of the line with y-intercept -6 and that is Parallel to the line with equation
$y = 3x/5 - 1$

⑤ Find an equation of the line that contains the point $(4, -4)$ and is perpendicular to the line with equation
$10(x-3) = y-2$.

⑥ Find an equation of the line with y-intercept $5/6$ and that is perpendicular to the line with equation $2x + 5y = -4$.

⑦ Convert the following equation to slope-intercept form.

$$y + 1 = 7(x-2)$$

Solve.

⑧ $4\sqrt{x+1} = \sqrt{x+2}$

⑨ $\dfrac{\sqrt{4-6x}}{2} + 9 = 4$

⑩ a) $1 = \sqrt{x^2 - 6x - 15}$

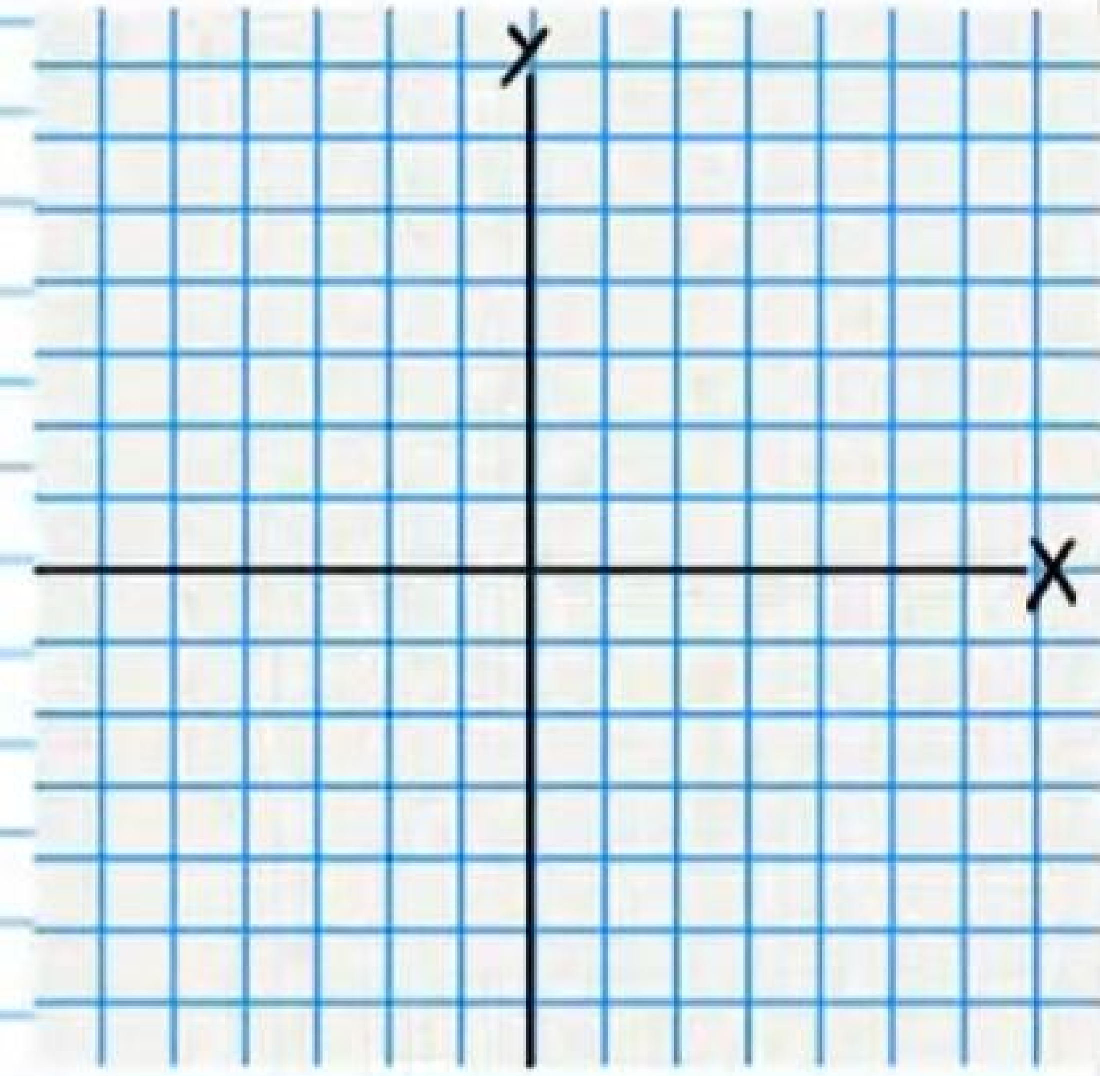
b) $\sqrt{7+3x} - 3 = x$

(11) $\frac{y^2}{11}+7=8$

(12) $46=\left(\frac{x}{4}+2\right)^2-3$

(13) a) Solve the following system using the graphical method.

$$\begin{cases} x+y=5 \\ 3x+5=y \end{cases}$$



b) Then solve the system algebraically (i.e. substitution or elimination) to check your solution.

$$\begin{cases} x+y=5 \\ 3x+5=y \end{cases}$$

Solve.

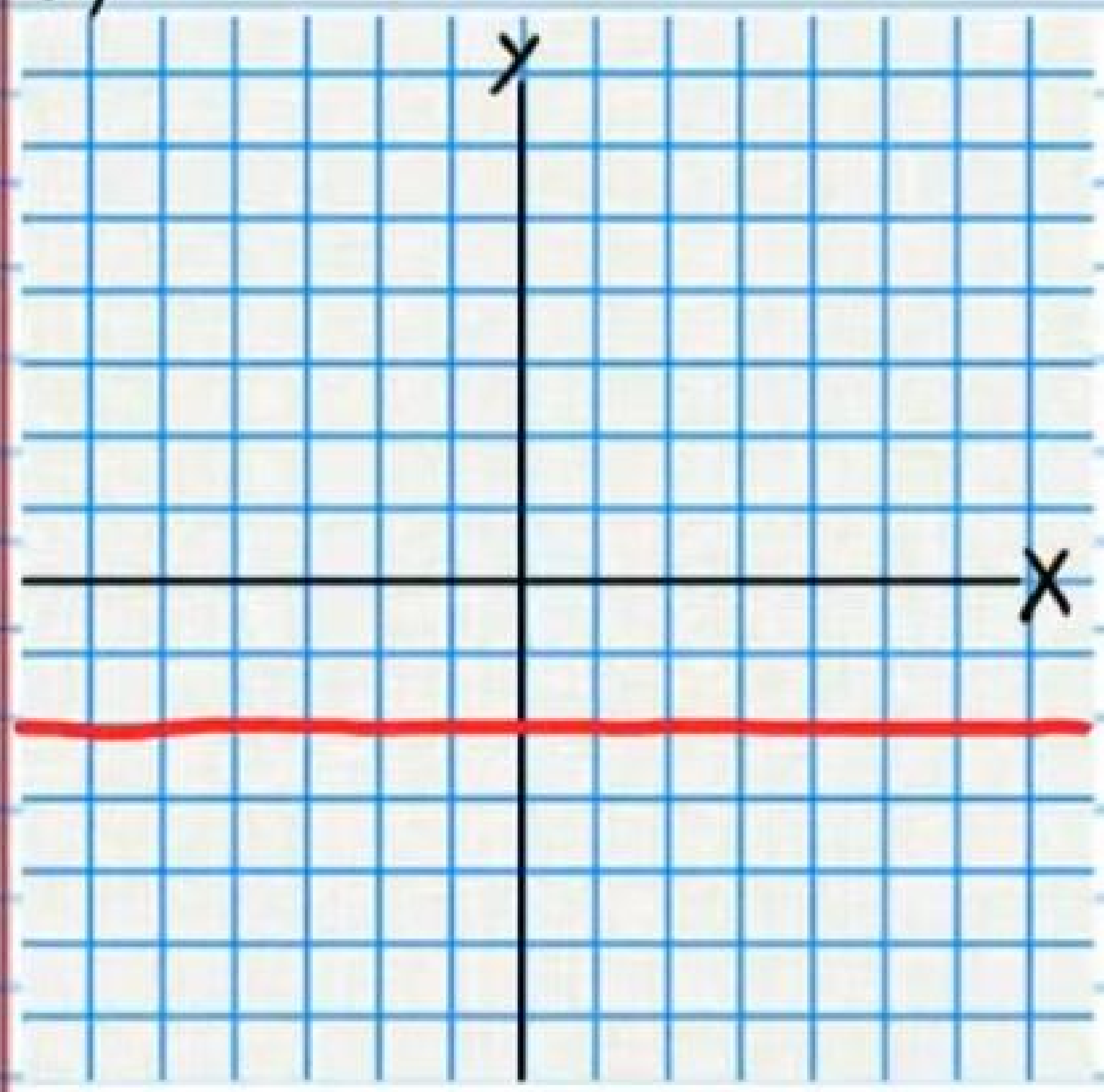
14 $\frac{4}{x^2}+\frac{1}{3x}=\frac{7}{2x}$

15 $\begin{cases} 2y+1=x \\ -3x+2y=6 \end{cases}$

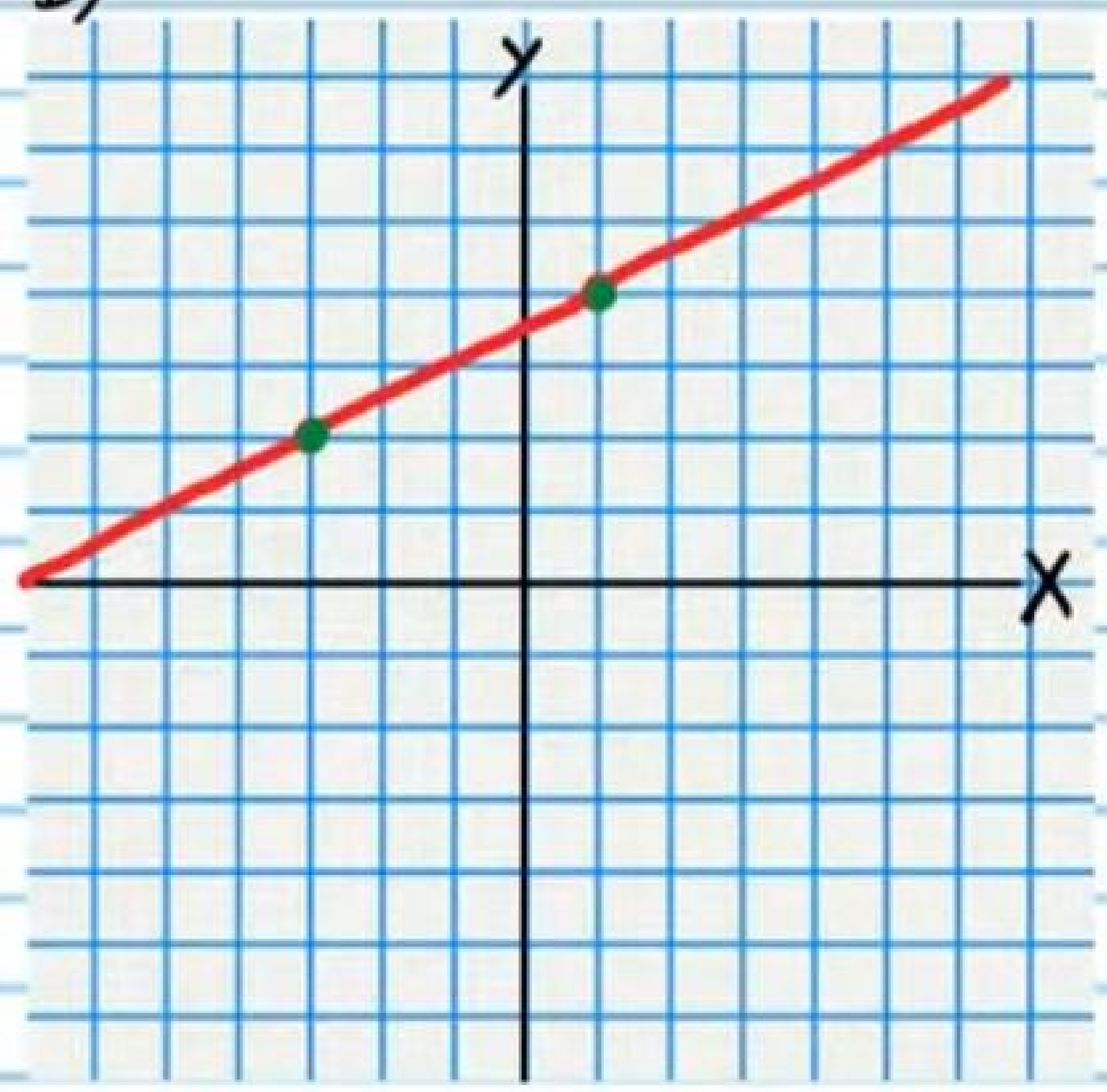
16 $\frac{1}{x-1}+\frac{15}{x^3+x^2-2x}=\frac{-5}{x^2+2x}$

⑰ For each graph below, write an equation of the line shown and identify the slope.

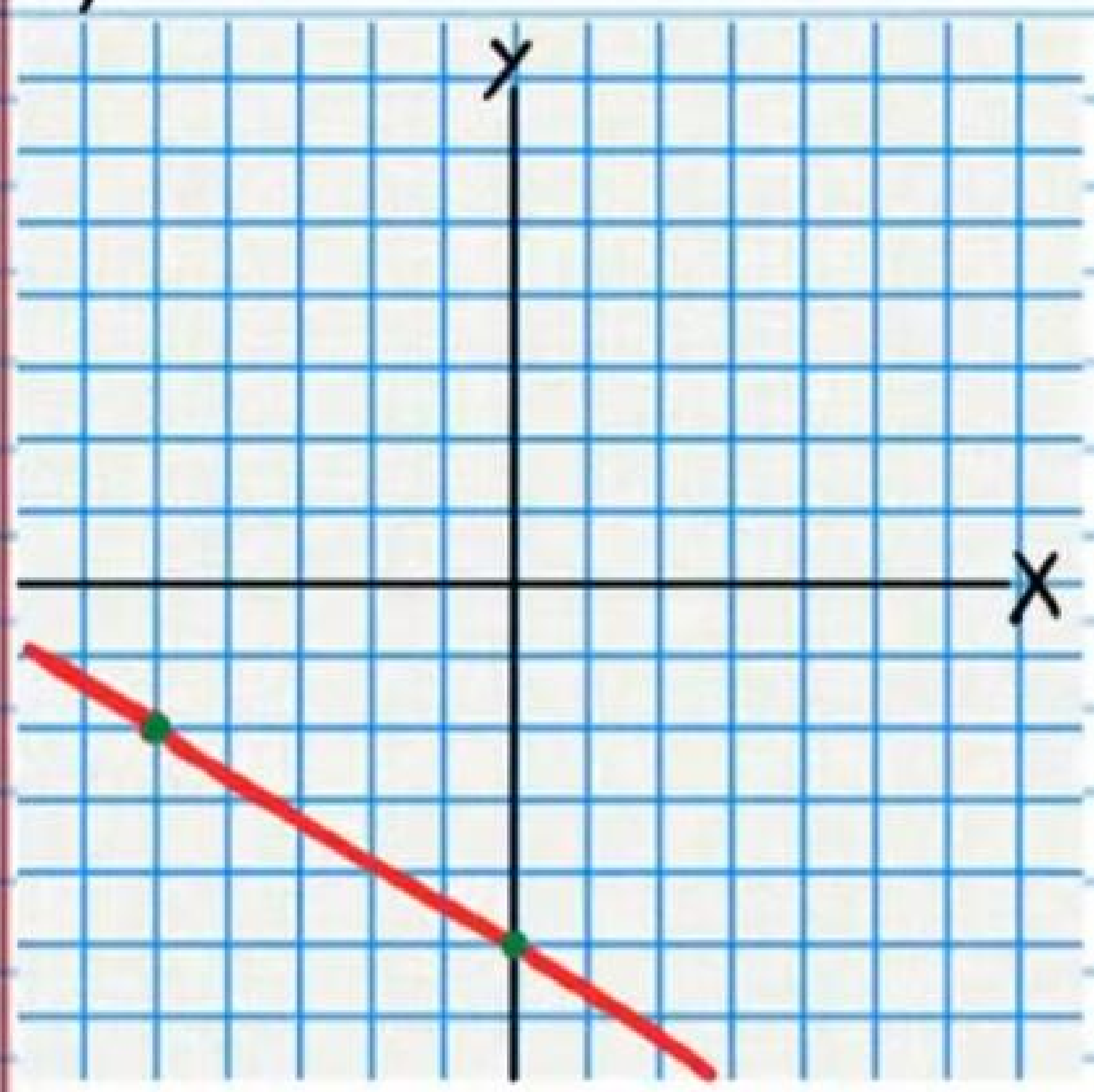
a)



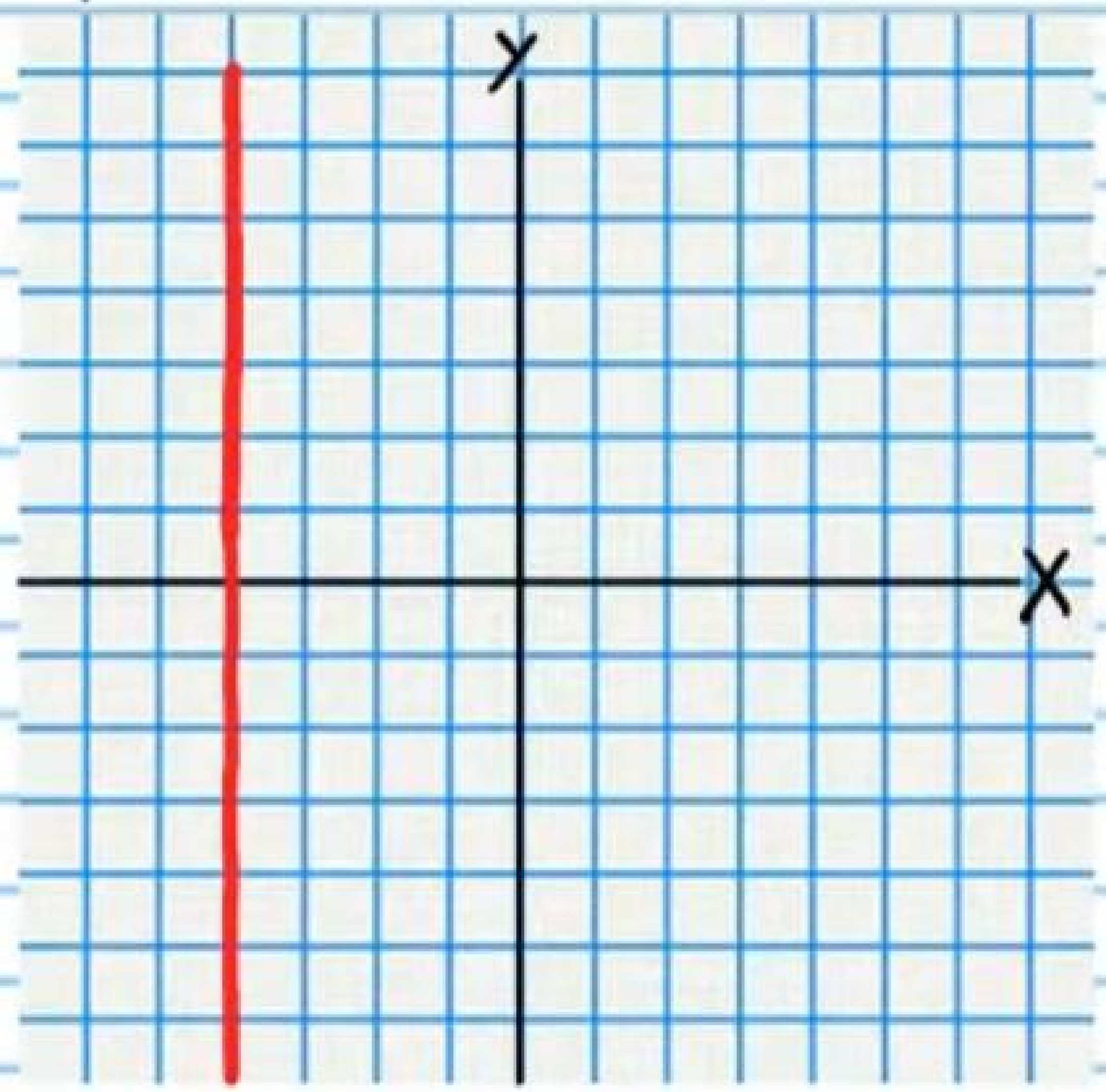
b)



c)



d)



⑱ Joseph bought 3 glazed donuts and 8 crumb donuts for \$18. At the same donut shop, Marvin bought 5 glazed donuts and 6 crumb donuts for \$19. How much does each type of donut cost?

⑲ Graph the following equations and identify the slopes. Label axes and intervals.

a) $x=0$

b) $y=2$

Solve.

(20) $$\frac{2x}{4x^2-13x-12}-\frac{3}{4x+3}=\frac{9}{4-x}$$

(21) $$\frac{x}{x-1}+1=\frac{4}{x-3}+\frac{x^2-5x+2}{x^2-4x+3}$$

(22) Nadia has 26 coins in her purse, only dimes and nickels. If the total value of these coins is \$2.15, how many nickels does she have?

(23) Lemonade is made by mixing concentrate with water. If lemonade is 8% concentrate, then that means that 8% of it is concentrate and the rest is water. If lemonade is 20% concentrate, then that means that 20% of it is concentrate and the rest is water. Kevin will mix an 8% concentrate lemonade with a 20% concentrate lemonade to make 7 quarts of a 12% concentrate lemonade. How many quarts of each premix should he combine?

(24) A cat leaps into the air. Its height (in feet) t seconds after the jump can be modeled by the equation below.

$$h = -16t^2 + 8t$$

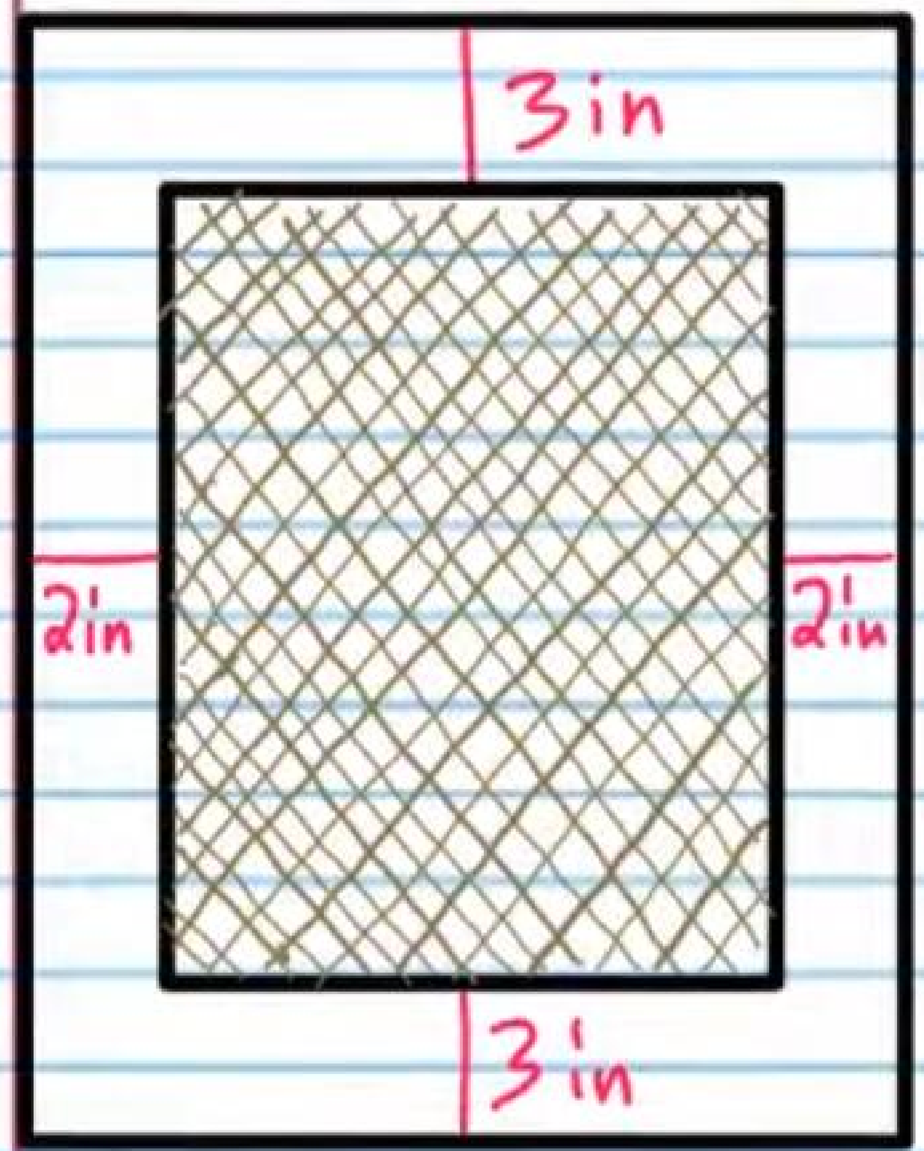
a) What is the height of the cat $\frac{1}{5}$ of a second after the jump?

b) How many seconds after jumping will the cat be one foot in the air?

c) How many seconds after jumping before the cat lands (Hint: there will be two answers to your equation, one that is relevant to this problem and one that is not)?

d) Explain in your own words why one of the answers for part "C" is not relevant. You must use complete sentences and proper grammar.

(25) The face of a heater has a flat rectangular shape and the inner mesh where the warm air comes out has a smaller rectangular shape. The area of the entire face (including the mesh) is 120 square inches more than the area of the mesh alone. If the length of the rectangular mesh is 4 more inches than its width, what are the dimensions of the entire heater?



Answers for Version 2 Test

① a) $\boxed{y-2=x+2}$ or $\boxed{y-2=1(x+2)}$ b) $\boxed{y=x+4}$

c) $\boxed{m=1}$ d) $\boxed{b=4}$

② $\boxed{y=\frac{2}{3}x+\frac{20}{3}}$

③ $\boxed{y=2(x-2)}$ or $\boxed{y+4=2(x-0)}$ or $\boxed{y=2x-4}$

④ $\boxed{y=\frac{3}{5}x-6}$

⑤ $\boxed{y+4=-\frac{1}{10}(x-4)}$ or $\boxed{y=-\frac{x}{10}-\frac{18}{5}}$

⑥ $\boxed{y=\frac{5}{2}x+\frac{5}{6}}$ or $\boxed{y-\frac{5}{6}=\frac{5}{2}(x-0)}$

⑦ $\boxed{y=7x-15}$

⑧ $\boxed{x=-\frac{14}{15}}$

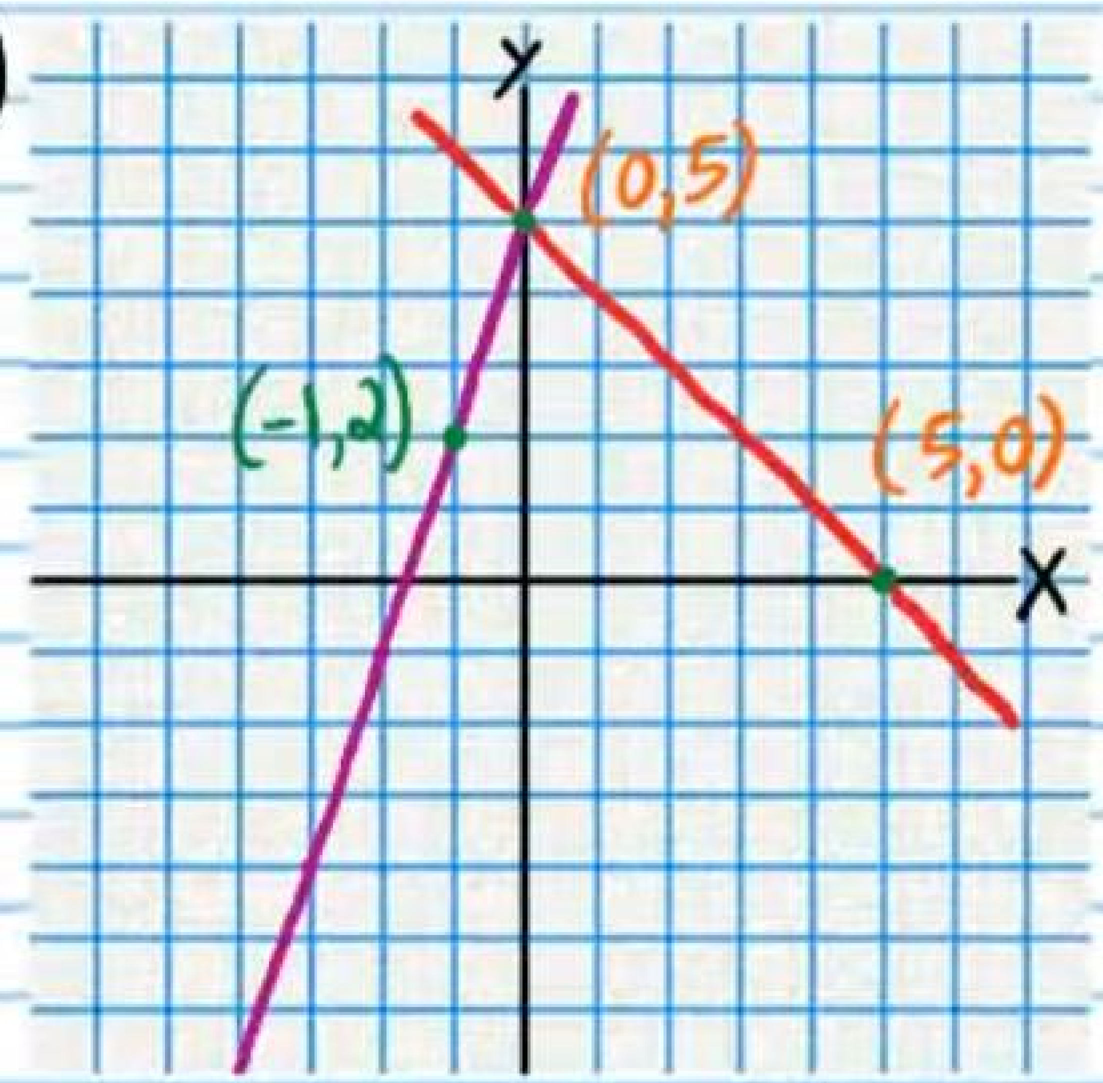
⑨ $\boxed{\text{No Solution}}$

⑩ a) $\boxed{x=8 \text{ or } x=-2}$ b) $\boxed{x=-2 \text{ or } x=-1}$

⑪ $\boxed{x=\pm\sqrt{11}}$

⑫ $\boxed{x=20 \text{ or } x=-36}$

⑬ a)



$\boxed{x=0,\ y=5}$ or $\boxed{(0,5)}$

b) Show your work: $\boxed{x=0,\ y=5}$ or $\boxed{(0,5)}$

⑭ $\boxed{x=\frac{24}{19}}$

⑮ $\boxed{\left(-\frac{7}{2},-\frac{9}{4}\right)}$

⑯ $\boxed{x=-5}$ or $x\neq-2$

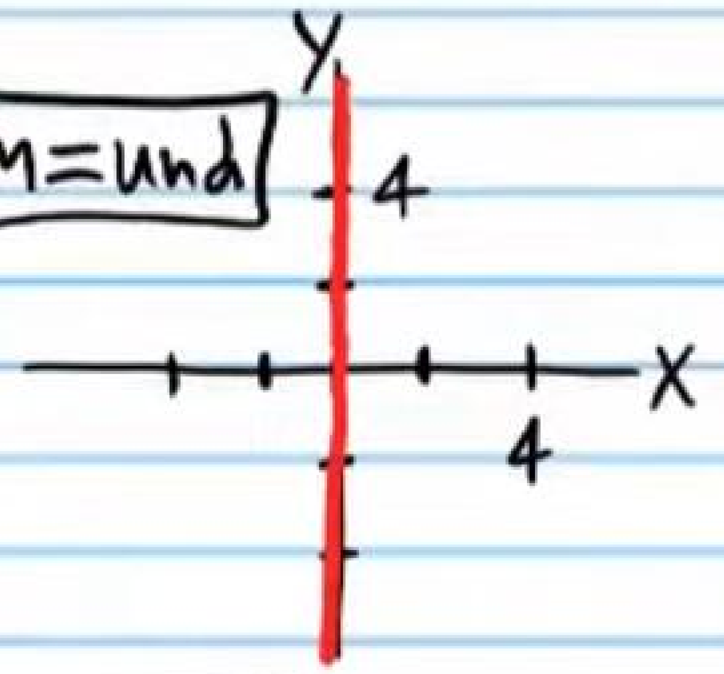
⑰ a) $\boxed{y=-2,\ m=0}$ b) $\boxed{y-4=\frac{1}{2}(x-1) \text{ or } y-2=\frac{1}{2}(x+3),\ m=\frac{1}{2}}$

c) $\boxed{y=-\frac{3}{5}x-5 \text{ or } y+2=-\frac{3}{5}(x+5) \text{ or } y+5=-\frac{3}{5}(x-0),\ m=-\frac{3}{5}}$

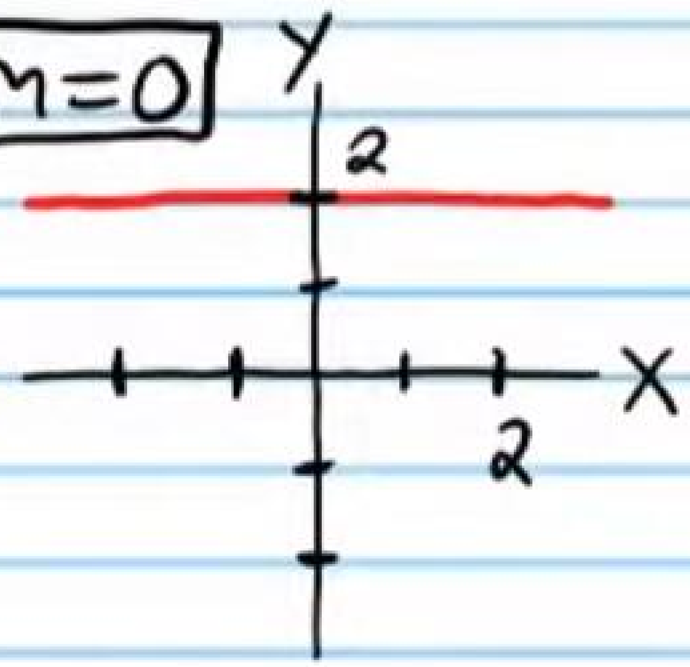
d) $\boxed{x=-4,\ m=\text{und}}$

(18) $\boxed{\$1.50/\text{crumb}, \ \$2/\text{glazed}}$

(19) a) $\boxed{m = \text{und}}$



b) $\boxed{m = 0}$



(20) $\boxed{x = -\frac{39}{35}}$

(21) $\boxed{x = 5}$, $x \neq 1$

(22) $\boxed{\text{9 Nickels}}$

(23) $\boxed{4.\overline{6} \text{ quarts of } 8\%, \ 2.\overline{3} \text{ quarts of } 20\%}$

$\boxed{\text{or } 4\tfrac{2}{3} \text{ quarts of } 8\%, \ 2\tfrac{1}{3} \text{ quarts of } 20\%}$

(24) a) $\boxed{\frac{24}{25} \text{ ft}}$ b) $\boxed{\frac{1}{4} \text{ s}}$ c) $\boxed{\frac{1}{2} \text{ s}}$

d) The cat is on the ground at $t = \frac{1}{2}$ and $t = 0$ but the second solution doesn't give us any information about how long the cat was in the air after the jump. It only tells us that the cat was on the ground when the clock started.

(25) $\boxed{W = 12 \text{ in}, \ L = 18 \text{ in}}$ or $\boxed{12 \text{ in by } 18 \text{ in}}$